

ANATOMY OF THE AR

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THE ARM

- Elbow joint
- proximal radio-ulnar joint
- Cubital fossa

ELBOW JOINT

ELBOW JOINT

❖ Type & Variety ;

- synovial of Hinge variety (uniaxial)
- Compound joint (3 bones)

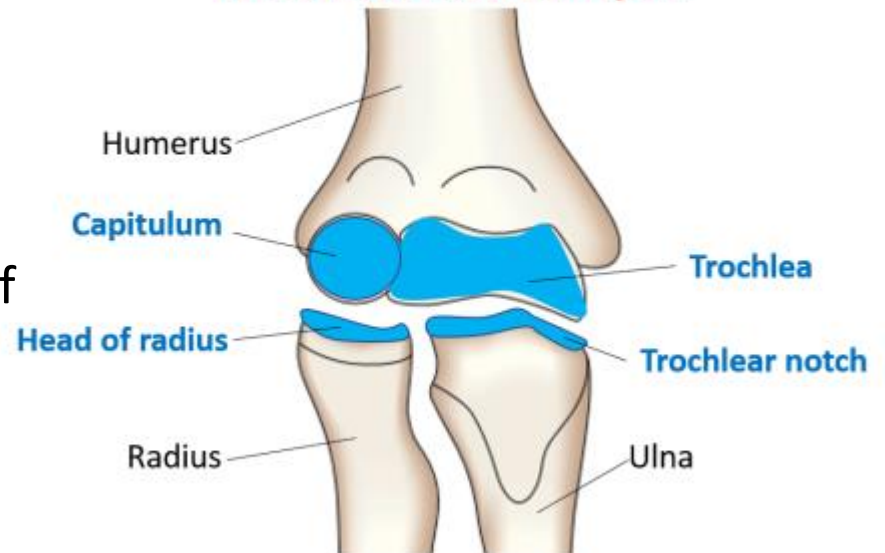
❖ Articular surfaces:

- Trochlea & capitulum of the lower end of the humerus
- Trochlear notch of ulna & head of radius

❖ Capsule:

- thin in front & behind
- thick at side

Articular surfaces of Elbow joint



Attachment of Capsule of Elbow joint



ELBOW JOINT

❖Ligament :

- as every hinge joint supported by 2 strong collateral ligament

1) radial collateral ligament

- ✓ Strong triangular ligament on the late side
- ✓ Apex attach to lat. Epicondyle
- ✓ Base to annular ligament

2) Ulnar collateral ligament

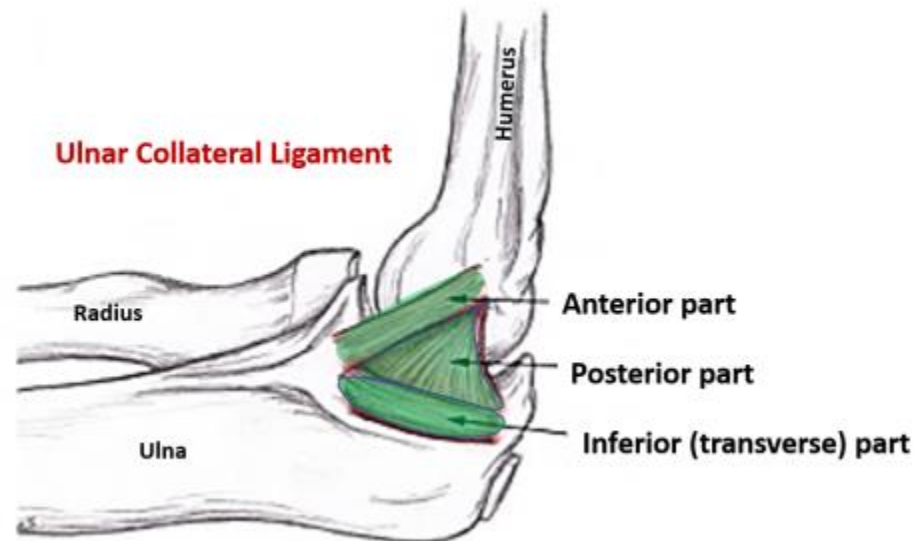
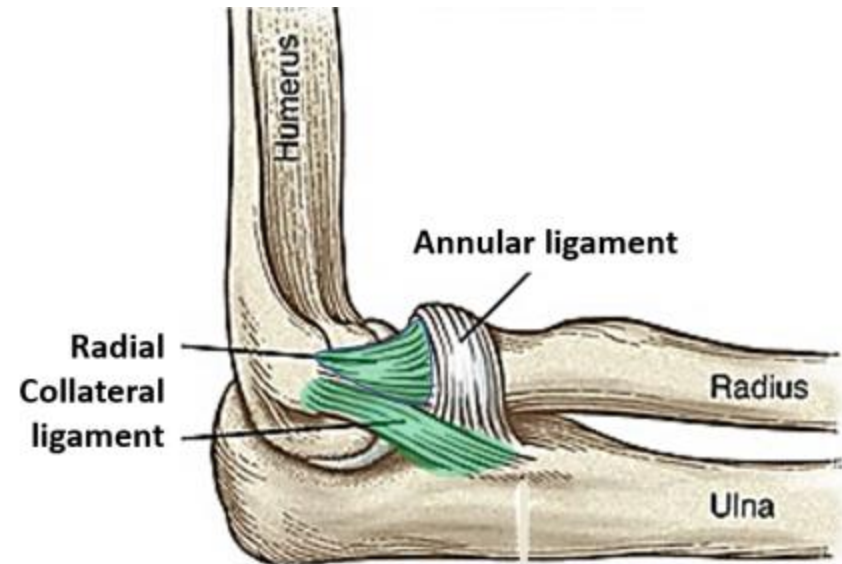
- ✓ Strong triangular ligament on the med. Side
- ✓ Apex attach to med.. Epicondyle
- ✓ Its ant. border : extend from med. Epicondyle to medial border of the coronoid process of ulna
- ✓ Post border extends from med. Epicondyle to med. Margin of olecranon process
- ✓ Lower border extends from coronoid process to olecranon process

N.B :

Ulnar collateral ligament is **crossed by ulnar n.**

❖Movement;

Flexion ,extension



Carrying angle

❖ Definition

- ★ is the angle between the long axes of the arm and forearm when the elbow is fully extended
- ★ It is about 15 degree .
- ★ It is wider in females

★ Importance

- ★ permits forearm to hip in swinging movement during walking and is important in carrying objects

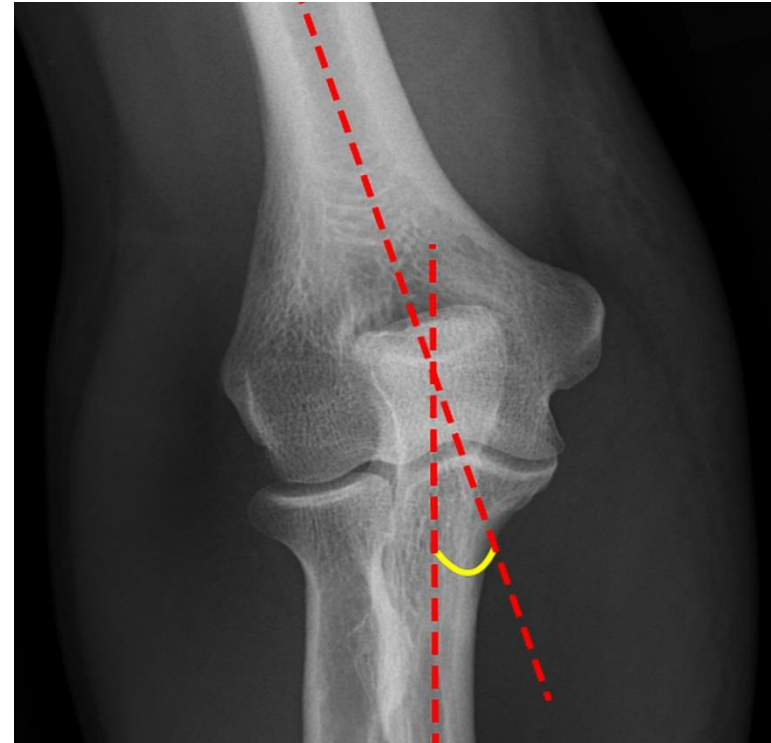
★ Abnormities



= Cubitus varus



= Cubitus valgus



Cubitus varus



Normal



Cubitus valgus

SUPERIOR(PROXIMAL) RADIOULNAR JOINT

proximal radio-ulnar joint

❖Type & Varity :

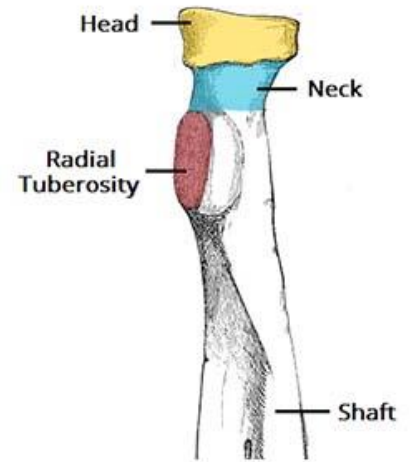
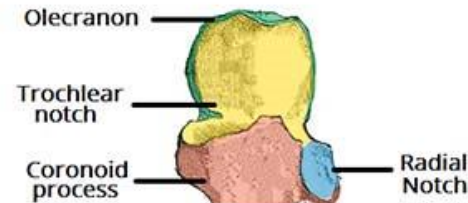
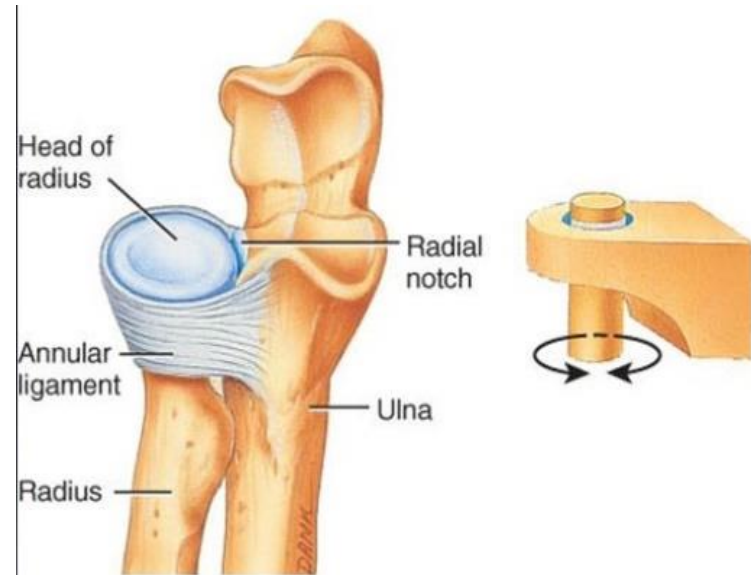
Synovial, uniaxial (pivot) joint.

Articular surfaces

- 1- head of radius
- 2- radial notch of ulna & annular ligament

❖Capsule;

- Covers the annular ligament
- Above Continuous with capsule of elbow joint. **Their joint spaces are continuous together**
- Below loosely attached to the neck of the radius
- The synovial membrane is a downward prolongation of the synovial membrane of the elbow joint



proximal radio-ulnar joint

❖ Ligaments :

1) annular ligament

- strong curved
- form $\frac{3}{4}$ of the circle With radial notch

2) Quadrate ligament

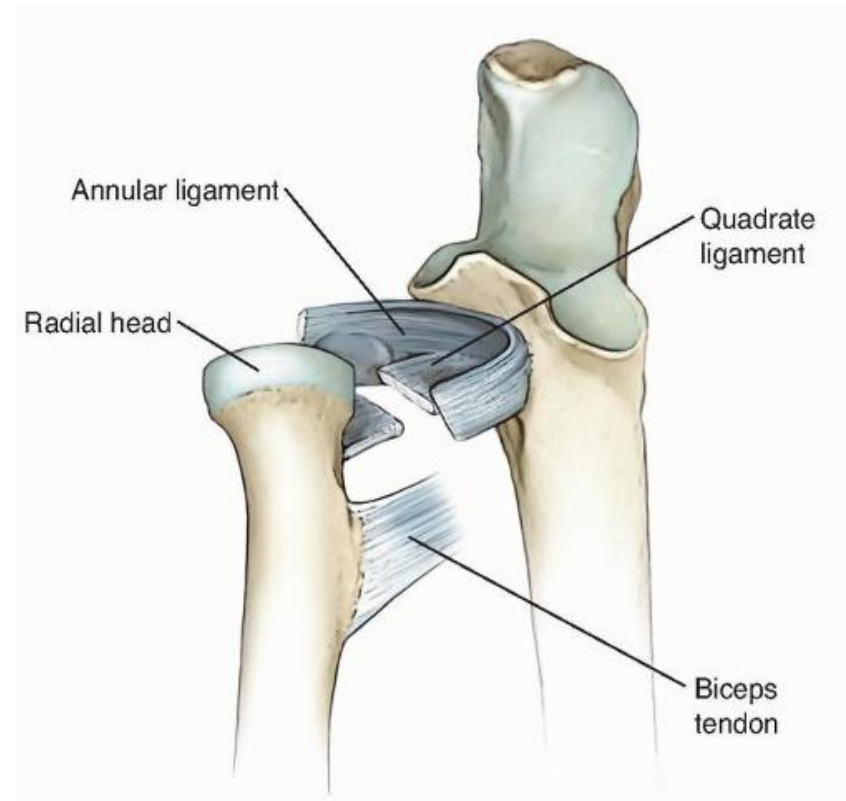
- thin quadrangular fibrous band
- extends from lower border of radial notch of ulna medially to the neck of radius laterally
- closing the joint infromedially

N.B:

cavity of proximal radioulnar joint is continuous with elbow joint cavity above

❖ Movement

- supination & pronation



1) It is initiated by brachioradialis (rotates the radius from full **SUPINATION** to midprone position).

2) It is completed by

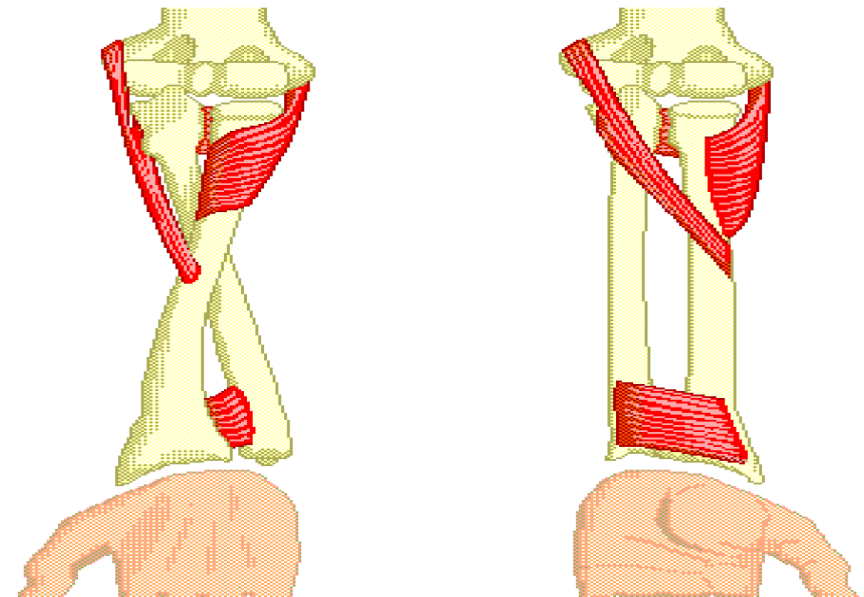
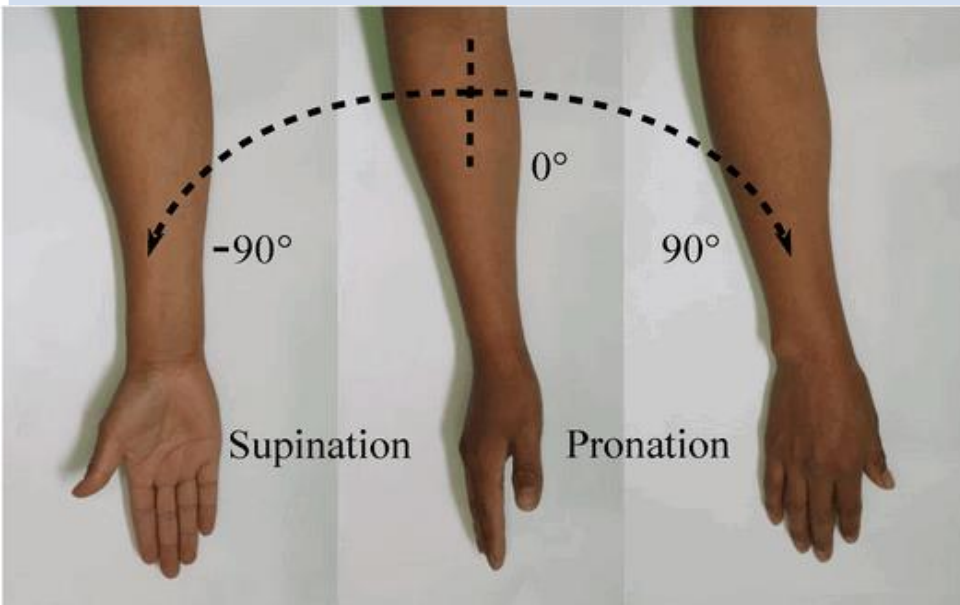
- Pronator teres
- pronator quadratus

1) It is initiated by brachioradialis (rotates the radius from full **PRONATION** to midprone position).

2) It is completed by

- supinator (when elbow is extended)
- Biceps (when elbow is flexed)

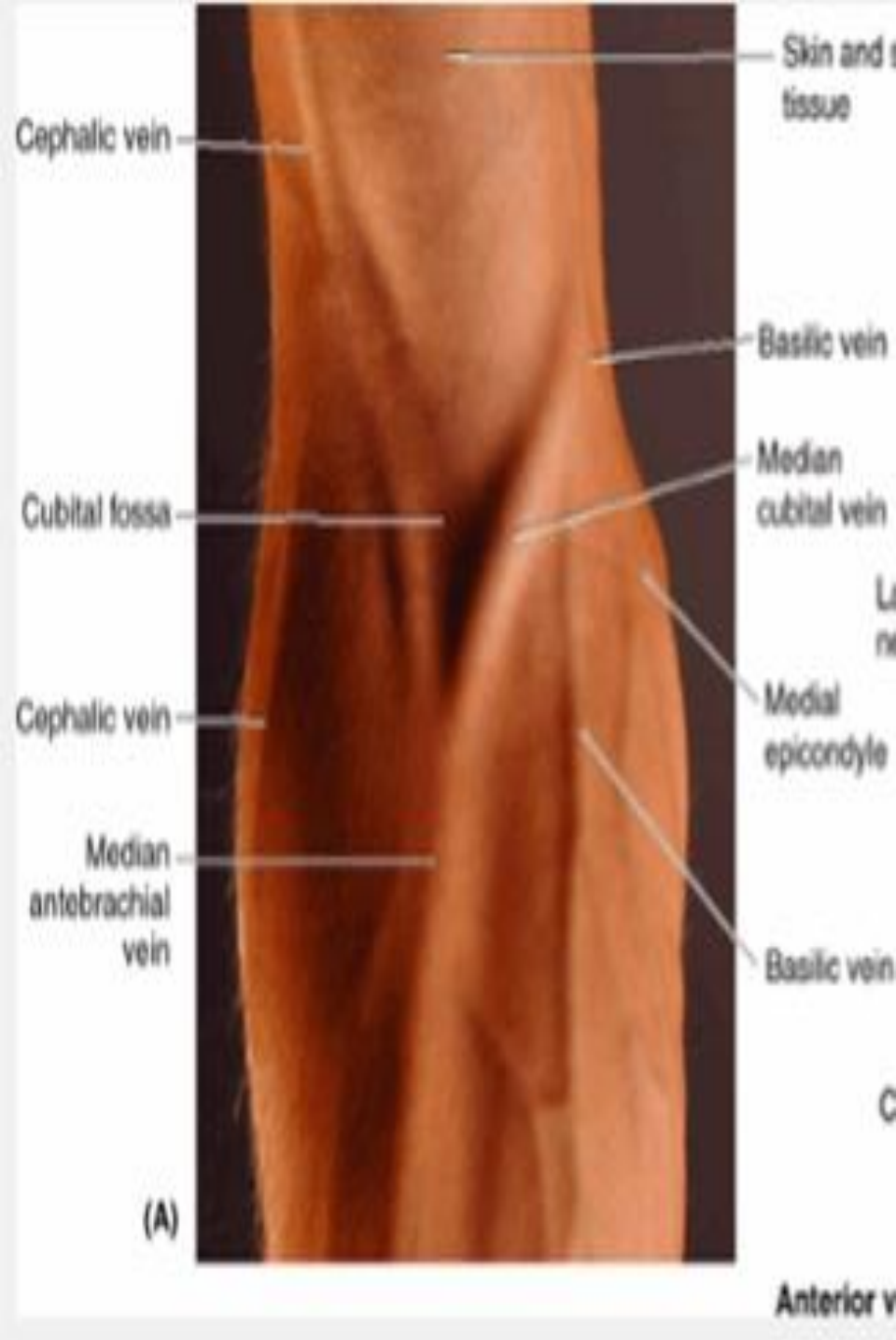
Axis of movement is ULNA



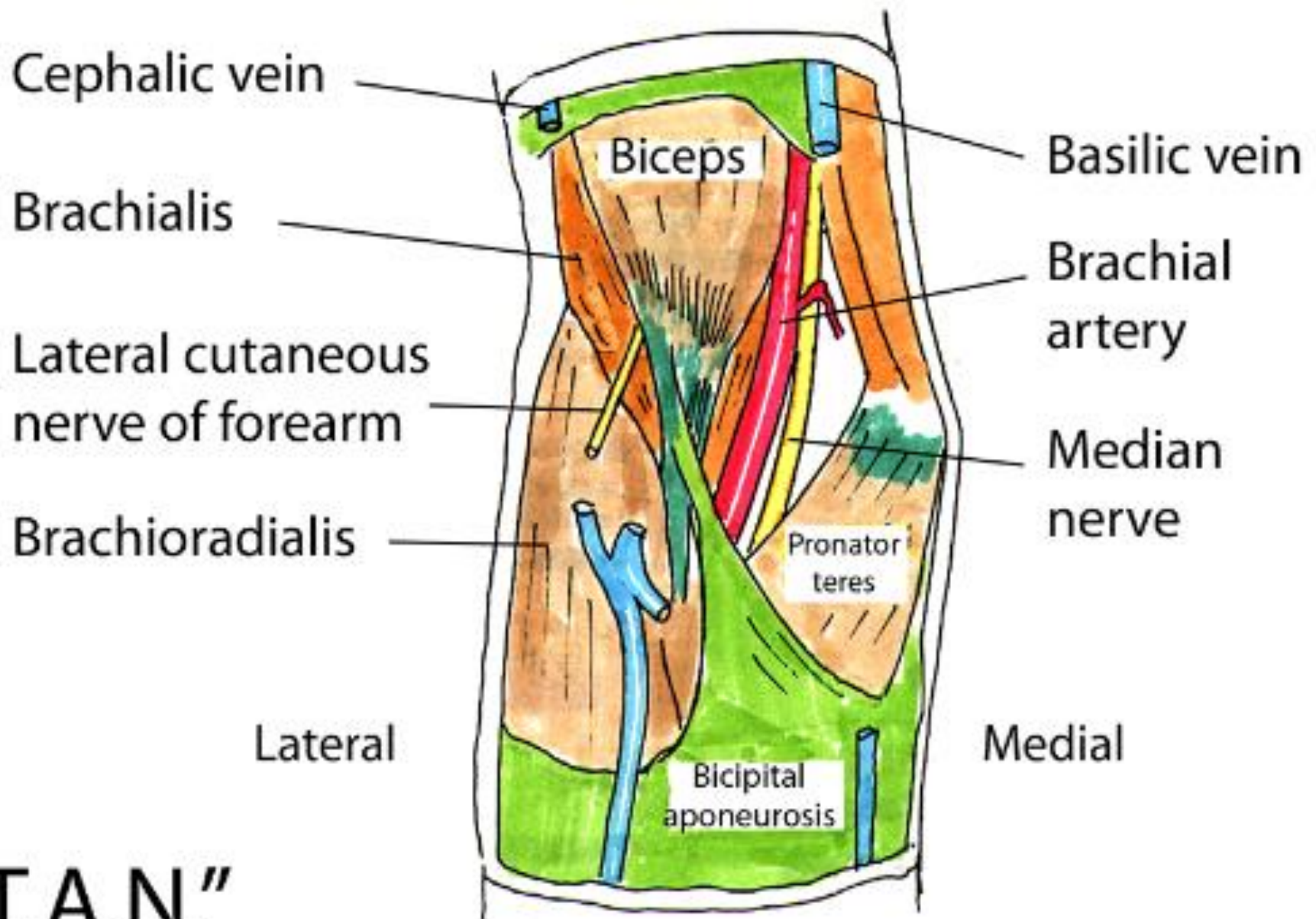
CUBITAL FOSSA

CUBITAL FOSSA

- Site
- Shape
- Boundaries
 - Medial Border
 - Lateral Border
 - Apex
 - Base
 - Floor
 - Roof
- Contents

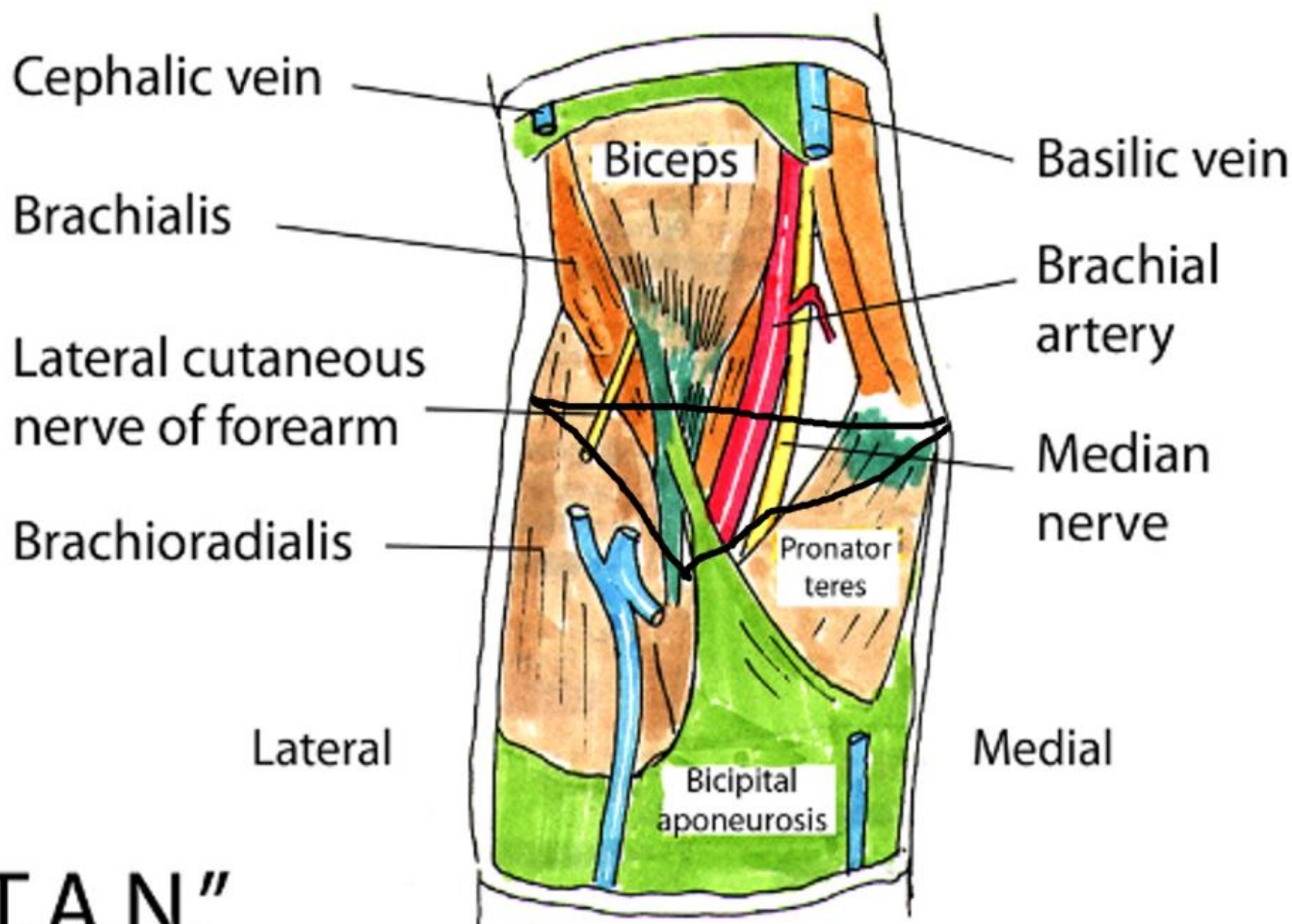


CUBITAL FOSSA



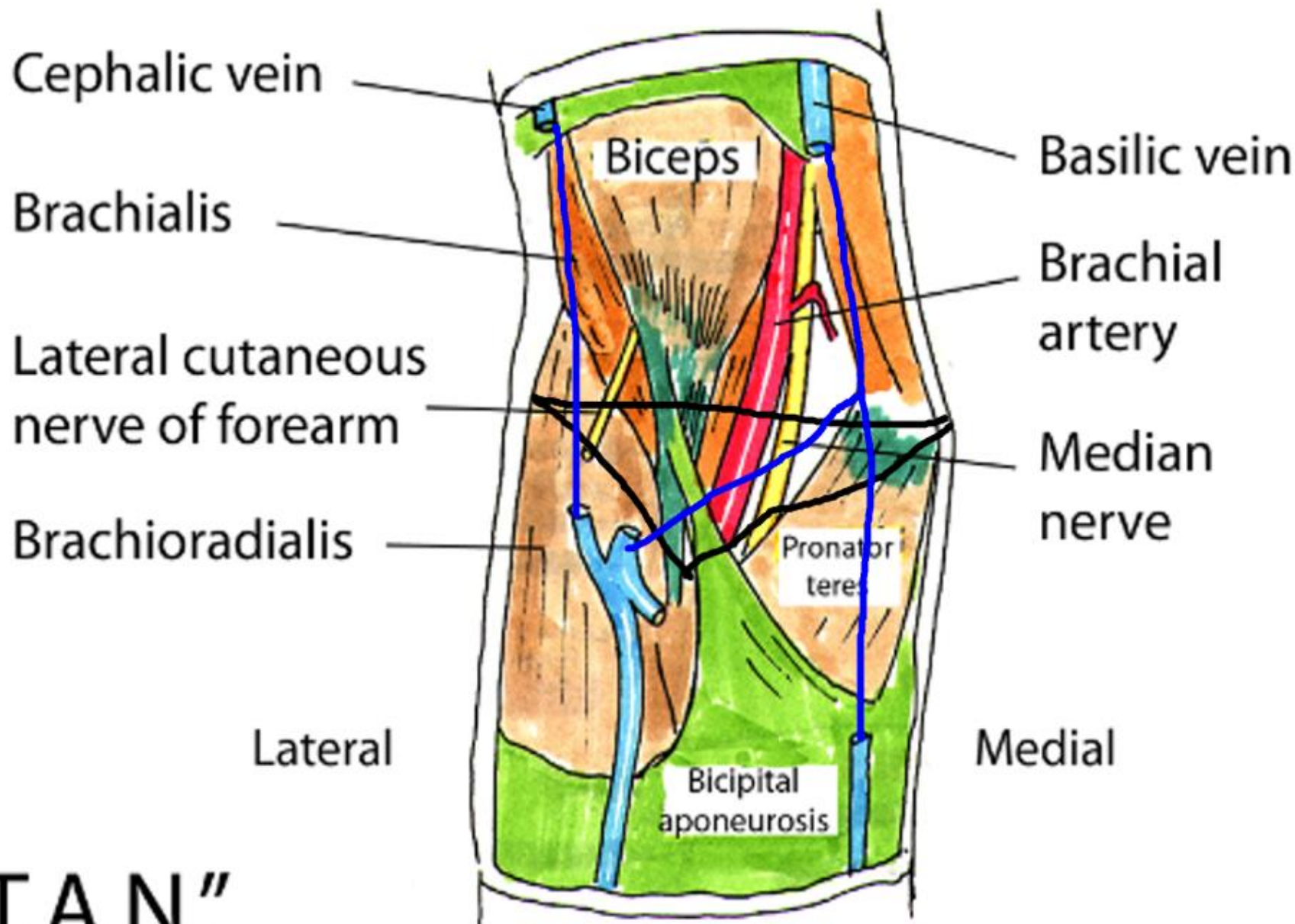
"T.A.N."

CUBITAL FOSSA

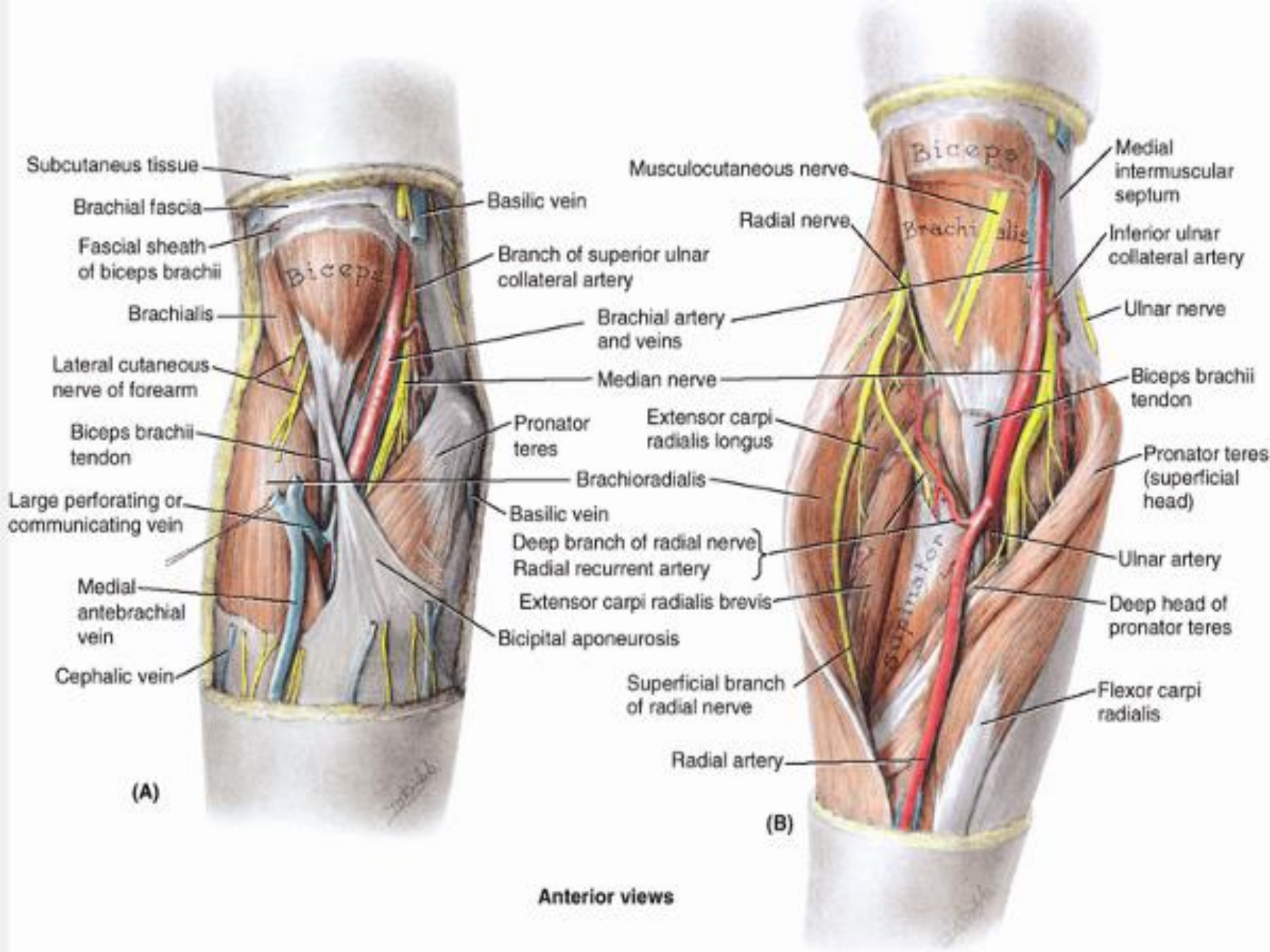


"T.A.N."

CUBITAL FOSSA



"T.A.N."



CUBITAL FOSSA

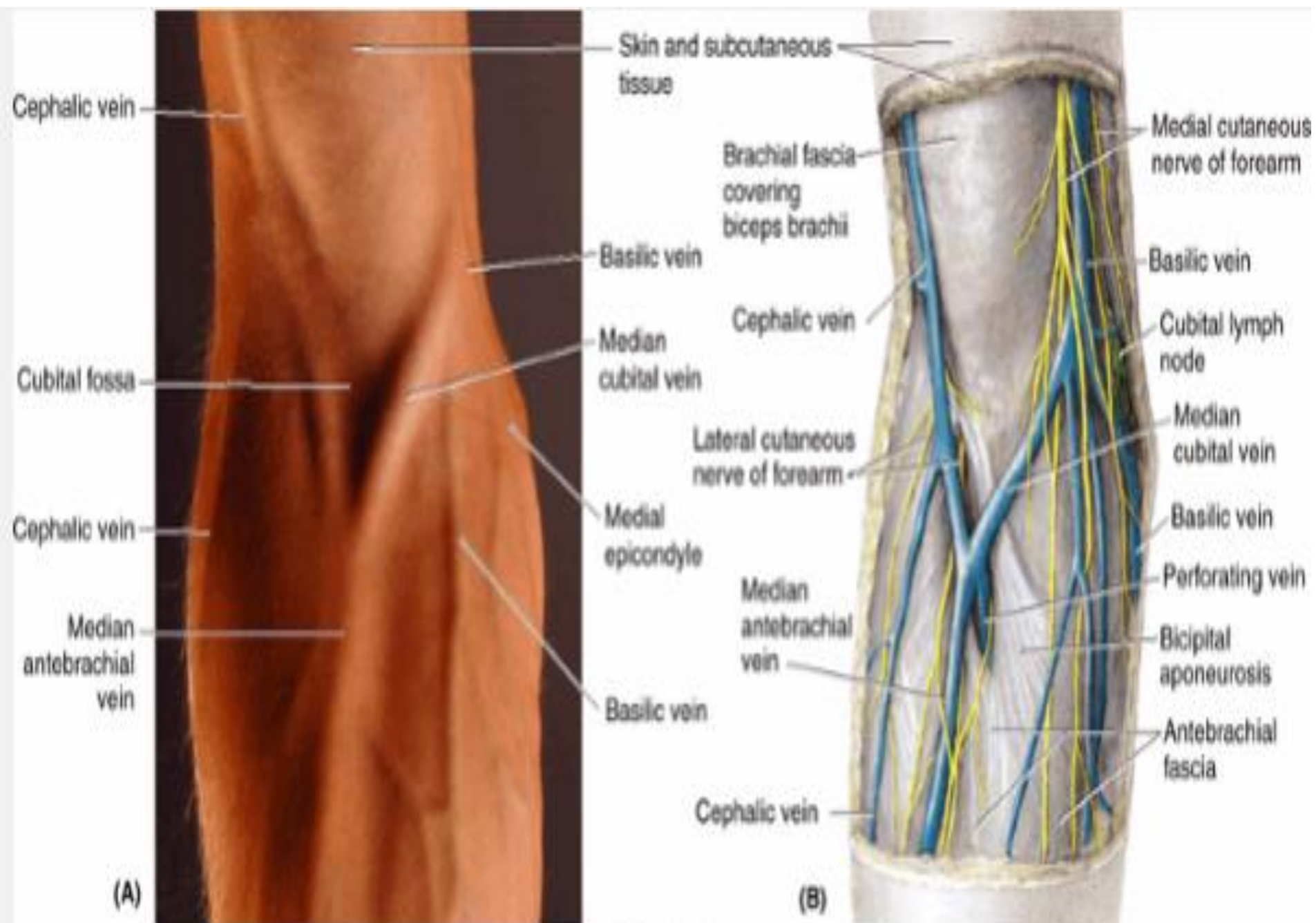
- **Definition & Site:** intermuscular space in front of elbow
- **Shape:** triangular in shape, its base above & apex below
- **Boundaries:**
 - ✓ **Base:** intercondylar line between to TWO humeral epicondyles
 - ✓ **Medial border** : pronator teres m.
 - ✓ **Lateral border:** brachioradialis m
 - ✓ **Apex:** formed by the meeting of the medial & lateral borders
 - ✓ **floor** : formed by
 - 1- brachialis (above & medially)
 - 2- supinator (below & laterally)
 - ✓ **Roof:** formed by
 - 1-skin
 - 2- superficial fascia containing
 - a) basilic v. with med. Cut. N. of forearm (medially)
 - b) cephalic v. lat. Cut. N. of forearm (laterally)
 - c) median cubital v. connecting the basilic & cephalic veins
 - 3- deep fascia (reinforced by the bicipital aponeurosis)

CUBITAL FOSSA

□ **Contents:** arranged from medial to lateral as follows:

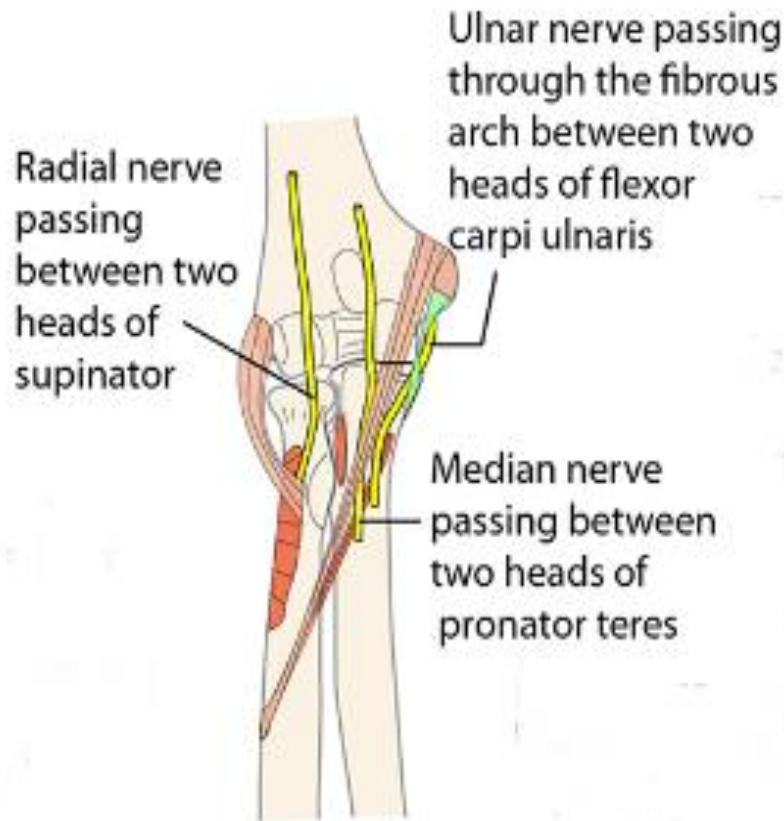
- 1- **median n.:** leaves the fossa by passing between the 2 heads of pronator teres
- 2- **the terminal part of brachial a.** & the beginning of both radial & ulnar arteries
 - a) radial a. leaves the fossa by passing superficial to pronator teres
 - b) ulnar a. leaves the fossa by passing deep to pronator teres m.
- 3- **biceps tendon** passes backwards to reach its insertion in the radial tuberosity
- 4- **radial n. & its post. Interosseous br.**
 - **Radial n.** leaves the fossa by passing deep to the brachioradialis m.
 - **Post. Interosseous n.** leaves the fossa by piercing the supinator m.

N.B: cubital fossa contains one LN (epitrochlear LN) lying in front of the med. epicondyle



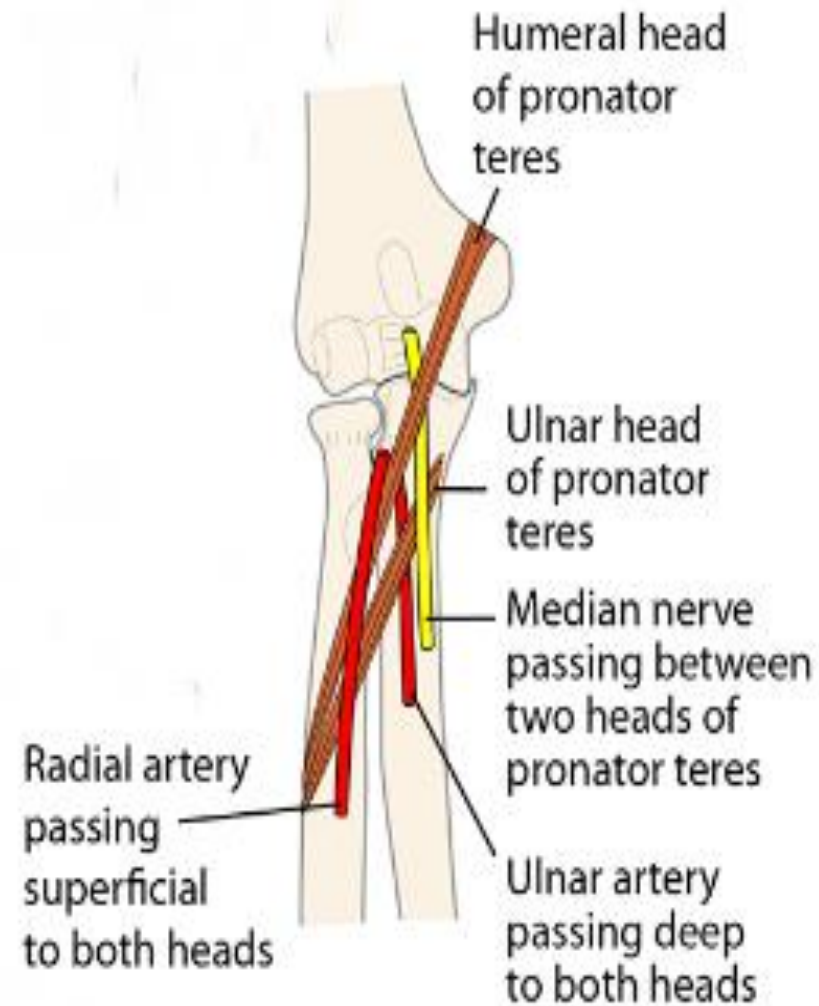
Anterior views

Relations of nerves & arteries in the elbow region



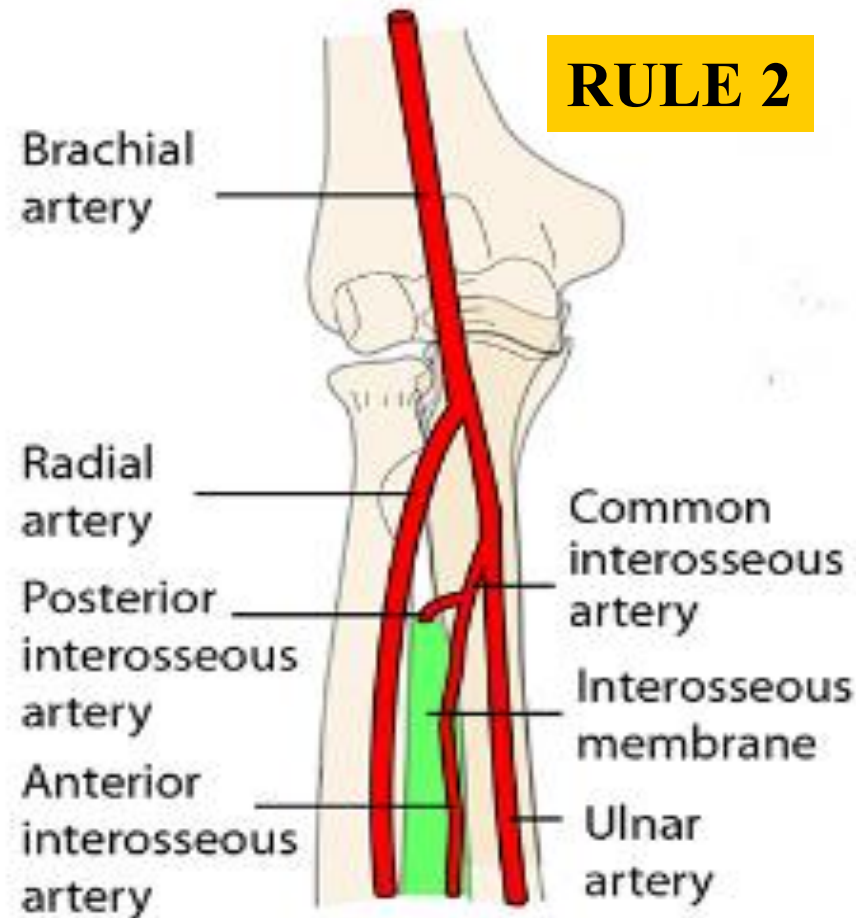
RULE 1

Nerves never pass between bones but pass around them.
Radial, median and ulnar nerves all pass between the two heads of muscles



Relations of the two heads of pronator teres

Relations of nerves & arteries in the elbow region



Arteries may pass between two bones as here where the posterior interosseous artery passes between ulna and radius